

* GS: $\theta^{\dot{\alpha}} \rightarrow \theta^{\dot{\beta}}$
 $\theta^{\dot{\alpha}} \rightarrow \theta^{\dot{\beta}} = \frac{m^{\dot{\beta}}}{p^+}$ | $d^{\dot{\alpha}} = p^{\dot{\alpha}} + P^+ \theta^{\dot{\alpha}} = 0$

$$d^{\dot{\alpha}} = p^{\dot{\alpha}} + (\gamma^m)^{\dot{\alpha}\dot{\beta}} p_m \theta_{\dot{\beta}} = p^{\dot{\alpha}} + P_+ (\gamma^+)^{\dot{\alpha}\dot{\beta}} \theta_{\dot{\beta}} + \dots$$

$$= p^{\dot{\alpha}} + P_- (\gamma^-)^{\dot{\alpha}\dot{\beta}} \theta_{\dot{\beta}}$$

$$= p^{\dot{\alpha}} + (-P^+)(\gamma^-)^{\dot{\alpha}\dot{\beta}} \theta_{\dot{\beta}}$$

$$\Rightarrow (\gamma^-)^{\dot{\alpha}\dot{\beta}} \theta_{\dot{\beta}} = -\theta^{\dot{\beta}}$$

$$d^{\dot{\alpha}} = p^{\dot{\alpha}} + P^+ \theta^{\dot{\alpha}} = 0$$

Pure-Spinor Super Particle =

$$S_{PS} = \int d\tau [\underline{P}_m \dot{x}^m + \omega_{\alpha} \dot{\lambda}^{\alpha} - P_{\alpha} \dot{\theta}^{\alpha}]$$

$$D=10: \quad \lambda \gamma^m \lambda = 0$$

$$\lambda^{\alpha} \lambda^{\beta} = (\gamma^{m_1 \dots m_{d/2}})^{\alpha\beta} (\lambda \gamma_{m_1 \dots m_{d/2}} \lambda)$$

~ Physical States:

$$Q_{BRST} = \lambda^{\alpha} \left(\underbrace{P_{\alpha} + \frac{1}{2} (\not{P} \theta)_{\alpha}}_{d_{\alpha}} \right)$$

$$= (\text{ghost})(\text{constraint \{second class\}})$$

* In RNS worldsheet: Took diffeo $f \mapsto \Lambda \lambda$
 susy $\epsilon \mapsto \Lambda \omega$

$$\text{ex: } \delta_f x^m = f \dot{x}^m \text{ (diffeo)}$$

$$\delta_{\epsilon} \chi^m = i \epsilon \gamma^m \chi^m \text{ (susy)}$$

$$\delta \chi^m = f \dot{x}^m + i \epsilon \gamma^m \chi^m$$

$$\longrightarrow = \Lambda \lambda \dot{x}^m + i (\Lambda \omega) \gamma^m \chi^m$$

(In RNS we have ghost fields (c, γ) , here we have (λ, ω))

* Question: does PS possess Lorentz, susy and
 Kappa Symmetry?

$$\delta \omega_{\alpha} = \Lambda^m (\gamma_m \lambda)_{\alpha}$$

* To solve $\lambda \gamma^m \lambda = 0$: From $SO(1,9) \rightarrow SO(10)$

$\{ SO(1,9) \rightarrow SO(5,5) \text{ in } D=10 \text{ similar to}$

$\{ SO(1,3) \rightarrow SO(2,2) \text{ in } D=4 \text{ for (real) Tensors}$

look at $SO(10)$ in terms of $U(5)$

$$1^+ \quad 1 \quad (\text{scalar}) =$$

$$\lambda_a \rightarrow \lambda_{ab} \quad \underline{10} \quad \square \quad \text{of } SU(5) \quad a, b = 1, \dots, 5 \quad \left(\frac{5(5-1)}{2}\right)$$

$$\gamma_m \rightarrow \gamma^a = \{\gamma^0 + i\gamma^1, \gamma^2 + i\gamma^3, \dots, \gamma^8 + i\gamma^9\}$$

(Complex rep)

$$\gamma^{\bar{a}} = \{\gamma^0 - i\gamma^1, \dots, \dots\}$$

Decompose spinor

$$\lambda^{\pm \pm \pm \pm \pm}: \text{eigenvalues of } (\gamma^0 \gamma^1, \gamma^2 \gamma^3, \gamma^4 \gamma^5, \gamma^6 \gamma^7, \gamma^8 \gamma^9)$$

$$\lambda^+ \equiv \lambda^{++++}$$

$$\lambda_{ab} \equiv \lambda^{++--}$$

$$\lambda^a \equiv \lambda^{+----}$$

Pure Spinor on patch where $\lambda^+ \neq 0$

$$\lambda \gamma^m \lambda = 0 \Rightarrow \lambda^a = \frac{1}{8} \epsilon^{abcd} \lambda_{bc} \lambda_{de}$$

$\Rightarrow$ 11 independent components:

$$\lambda^a \in \frac{SO(10)}{U(5)} \times \mathbb{C}^*, \quad \frac{|SO(10)|}{|U(5)|} = \frac{45}{25} \sim 20 \text{ real or } 10 \mathbb{C}$$

$$\Rightarrow \frac{|SO(10)|}{|U(5)|} + |\mathbb{C}| = \frac{11}{1}$$

projective

pure-spinor

* In $D=4$: λ^a : Weyl spinors are Pure

$$\gamma^0 + i\gamma^1 \lambda^{+\pm \dots \pm} = 0$$

$$(\gamma^0 + i\gamma^1) \lambda^{-\pm \dots \pm} = \lambda^{+\pm \dots \pm}$$

$$\gamma^0 + i\gamma^1 \sim \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\omega_a \rightarrow \omega_a \quad \underline{5} \quad \omega^{++++}$$

$$\omega_{ab} \rightarrow \omega_{ab} \quad \underline{10} \quad \omega^{ab} \sim \omega^{++--}$$

$SU(5)$

$$\nabla \omega_r \quad \underline{1} \equiv \omega^{----}$$

$V = \lambda^\alpha A_\alpha(x, \theta)$: Cohomology of Q with $gh \neq 1$

$$\omega_a : gh = -1$$

$$\lambda^\alpha : gh = 1$$

$$D_\beta = \frac{2}{g\theta^\alpha} + \frac{1}{2}(\gamma^m \theta)_\beta \frac{\partial}{\partial x^m}$$

$$QV = \lambda^\alpha \lambda^\beta D_\beta A_\alpha = 0 \quad (\text{10d susy derivative})$$

$$\delta V = Q\Omega \quad \Omega : \text{ghost} \neq 0$$

$$\delta V = \lambda^\alpha D_\alpha \Omega$$

$$\lambda^\alpha \lambda^\beta D_\alpha A_\beta = 0 \Rightarrow (\gamma_{m_1 \dots m_5}^{\alpha\beta}) D_\alpha A_\beta = 0$$

$$\delta A_\alpha = D_\alpha \Omega$$

$$A(x, \theta) = A_\alpha(x) + a_m(x) (\gamma^m \theta)_\alpha - \frac{1}{2} \gamma^\beta (\gamma^m \theta)_\beta (\gamma_m \theta)_\alpha$$

$$\text{Choose } \Omega = 1 + \theta^\alpha \gamma_\alpha + \dots$$

$$\Rightarrow \partial_m \partial^m a^n = 0 \quad (\text{on-shell})$$

$$\partial_m (\gamma^m \xi)_\alpha = 0 \quad (D=10 \text{ SYM})$$

* θ^α : World line scalar

$$\text{Twistor-line constraint: } P^m (\gamma_m \lambda)_\alpha = 0$$

$$D=4: P^m = \lambda^\alpha \lambda^{\dot{\alpha}} (\sigma^m)_{\alpha\dot{\alpha}}$$

$$\Rightarrow P\lambda = 0$$

$$(\bar{\sigma}_m)^{\dot{\alpha}\alpha} (\lambda^\beta \lambda^{\dot{\beta}} (\sigma^m)_{\beta\dot{\beta}}) \lambda_\alpha = (2 \delta_{\dot{\beta}}^{\dot{\alpha}} \delta_\beta^\alpha) \lambda^\beta \lambda^{\dot{\beta}} \lambda_\alpha$$

$$= 2 \lambda^\alpha \lambda_\alpha \lambda^{\dot{\alpha}} = 0 \quad \text{since } \lambda^\alpha \lambda_\alpha = 0$$

$$\text{Note this is simply } P^m (\gamma_m \lambda)_\alpha = 0$$

$$(P\lambda)_\alpha = 0 \Rightarrow P(P\lambda)_\alpha = P^2 \lambda_\alpha = 0$$

$$\text{or } P^2 = 0 \quad \lambda^\alpha \in \mathbb{C}^4 \quad \text{so cannot vanish}$$

λ^α : projective pure spinor

$$S_c = \int d\tau \left[P_m \dot{x}^m - \omega_\alpha \nabla \lambda^\alpha - \frac{1}{2} \lambda^\alpha (P\lambda)_\alpha \right]$$

$\hat{=}$ Lagrange multiplier

$$\text{Scale-symm: } \nabla \lambda^\alpha = \dot{\lambda}^\alpha + A \lambda^\alpha$$

A : World line gauge field

$$\lambda^\alpha \rightarrow \Omega \lambda^\alpha$$

$$A \rightarrow A - \Omega^{-1} \dot{\Omega}$$

$$\omega_\alpha \rightarrow \omega_\alpha^{-1}$$

$$v^m \rightarrow v^m$$

$$\begin{cases} \omega_a \rightarrow \Omega^{-1} \omega_a \\ L^\alpha \rightarrow \Omega^{-1} L^\alpha \\ P_m \rightarrow 0 \end{cases}$$

$$\begin{aligned} \Rightarrow \delta S &= \int d\tau \delta \left[P_m \dot{x}^m - \omega_a \nabla \lambda^\alpha - \frac{1}{2} L^\alpha (\not{P} \lambda)_\alpha \right] \\ &= \int d\tau \left(-\delta \omega_a \dot{\lambda}^\alpha - \omega_a \delta (\dot{\lambda}^\alpha + A \lambda^\alpha) - \frac{1}{2} \delta L^\alpha (\not{P} \lambda)_\alpha \right. \\ &\quad \left. - \frac{1}{2} L^\alpha (\not{P} \delta \lambda)_\alpha \right) \\ &= \int d\tau \left(-\Omega^{-1} \omega_a \dot{\lambda}^\alpha - \omega_a (\Omega \lambda^\alpha + (A - \Omega^{-1} \dot{\Omega}) \lambda^\alpha + A \Omega \lambda^\alpha) \right. \\ &\quad \left. - \frac{1}{2} \Omega^{-1} L^\alpha (\not{P} \lambda)_\alpha - \frac{1}{2} L^\alpha (\not{P} \Omega \lambda)_\alpha \right) \end{aligned}$$

$$\begin{aligned} \rightarrow \nabla' \lambda'^\alpha &= (\dot{\lambda}^\alpha + A' \lambda^\alpha) = (\dot{\Omega} \lambda^\alpha + \Omega \dot{\lambda}^\alpha + (A - \Omega^{-1} \dot{\Omega}) \Omega \lambda^\alpha) \\ &= (\dot{\Omega} \lambda^\alpha + \Omega \dot{\lambda}^\alpha + A \Omega \lambda^\alpha - \dot{\Omega} \lambda^\alpha) \\ &= \Omega (\dot{\lambda}^\alpha + A \lambda^\alpha) = \Omega \nabla \lambda^\alpha \end{aligned}$$

$$L^\alpha (\not{P} \lambda)_\alpha \rightarrow \Omega^{-1} L^\alpha (\not{P} \Omega \lambda)_\alpha = L^\alpha (\not{P} \lambda)_\alpha$$

* Scale-symmetry only acts on α variables and leaves spacetime invar.

$\rightarrow$ Pure Spinors: $D=4$ twistors

$\rightarrow$ 10D SYM $\rightarrow (D=4, N=4 \text{ SYM})$ / comes from first class constraints

Gauge Symmetry:

Θ^α : ghost field

$$\delta L^\alpha = \nabla \Theta^\alpha - \Theta \lambda^\alpha, \quad \delta x^m = \frac{1}{2} \lambda \gamma^m \Theta$$

$$\delta \omega_\alpha = -\frac{1}{2} (\not{P} \Theta)_\alpha$$

* Suppose λ^α is constant, $\lambda^\alpha = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$: $\lambda_{\text{cub}} = 0$
 $\lambda_a = 0$

$$(\not{P} \lambda)_\alpha = P_m (\gamma^m \lambda)_\alpha = 0$$

$$P_m = \begin{pmatrix} P_a \\ P_{\bar{a}} \end{pmatrix} \quad P_a \equiv (P^0 + i P^1, P^2 + i P^3, \dots)$$

$$\Rightarrow P_a (\gamma^a \lambda)_\alpha + P_{\bar{a}} (\gamma^{\bar{a}} \lambda)_\alpha = 0$$

$$\hookrightarrow P_a = 0 \text{ (constant)}$$

* For the topological string this is equivalent to

$$\partial x^a = 0 \mid \lambda^a = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow S_c = \int d\tau \left(P_A \dot{x}^A + P_{\bar{A}} \dot{x}^{\bar{A}} - \frac{1}{2} L^A P_A \right) \Big|_{\text{Action}}^{\text{top-}}$$

Parametrizes how complex structure is chosen

Topological String: fix the complex structure λ
 * Start w top string: integrate over the complex structures
 $\Rightarrow$ Gives you the Superstring

Gauge - for - Gauge: $\delta' \theta^\alpha = \varphi \lambda^\alpha$, $\delta' \rho = \nabla \varphi$

Pure spinor constraint $\lambda \gamma^\mu \lambda = 0 \Rightarrow \omega_\alpha \sim \omega_\alpha + \lambda^\mu (\gamma_\mu \lambda)_\alpha$
 φ : 6-f-B parameter
 $\delta' (\delta L) = \varphi \nabla \lambda^\alpha$
 $\delta L^\alpha = -\varphi \frac{\delta \chi}{\delta L^\alpha}$

χ : 6-fermion fermion.

φ : Boson

$$S = S_c + \int d\tau Q \chi$$

$$\delta' (Q \chi) = \frac{\delta \chi}{\delta L^\alpha} \varphi \nabla \lambda^\alpha$$

$$Q L^\alpha = \nabla \theta^\alpha - \rho \lambda^\alpha, \quad \Theta \chi^\mu = \frac{1}{2} \lambda \gamma^\mu \theta, \quad Q \theta^\alpha = \varphi \lambda^\alpha$$

$$Q \omega_\alpha = \frac{1}{2} (\rho \theta)_\alpha - \varphi \frac{\delta \chi}{\delta L^\alpha}, \quad Q \rho = \nabla \varphi$$

Scale symm.: $\theta^\alpha \rightarrow \Omega^{-1} \theta^\alpha$, $\varphi \rightarrow \Omega^{-2} \varphi$

PS Choose χ to fix L^α to zero

$$\chi = p_\alpha L^\alpha + \beta \rho \quad \left[\begin{array}{l} \uparrow \\ \text{anti-ghost} \end{array} \right]$$

• Gauge ρ to zero.

$$Q p_\alpha = M_\alpha, \quad Q \beta = N \Leftrightarrow Q M_\alpha = Q N = 0$$

(M_α, N) : "N-L fields".

$$Q \chi = M_\alpha L^\alpha + p_\alpha (\nabla \theta^\alpha - \rho \lambda^\alpha) + N \rho + \beta \nabla \varphi$$

E.o.m M_α : $L^\alpha = 0$, E.o.m of N : $\rho = 0$

$$S_{\text{on-shell}} = S_c + \int d\tau Q \chi = \int d\tau [p^\mu \dot{x}_\mu + \omega_\alpha \dot{\lambda}^\alpha - p_\alpha \dot{\theta}^\alpha + \beta \nabla \varphi + \omega_\alpha \nabla \lambda^\alpha - p_\alpha \nabla \theta^\alpha]$$

$$S = \int d\tau (p^\mu \dot{x}_\mu + \omega_\alpha \dot{\lambda}^\alpha + \omega_\alpha A \lambda^\alpha + p_\alpha \dot{\theta}^\alpha + p_\alpha A \theta^\alpha + \beta \dot{\varphi} + \beta A \varphi)$$

$\Rightarrow$ From this result:

$$S = \int d\tau (P^m \dot{x}_m + \omega_\alpha \dot{\lambda}^\alpha - p_\alpha \dot{\theta}^\alpha + \beta \dot{\psi} + A [\omega_\alpha \lambda^\alpha + p_\alpha \theta^\alpha - 2\beta \psi])$$

This implies

$$\begin{cases} \nabla \theta^\alpha = \dot{\theta}^\alpha - A \theta^\alpha \\ \nabla \psi = \dot{\psi} - 2A \psi \end{cases}$$

* Use Scale symmetry of $\psi \rightarrow \Omega^{-2} \psi$ to gauge $\psi = 1$ (Assume $\psi(\tau) \neq 0 \forall \tau$ on worldline)

→ Note that the Scale determines these values

$$\begin{aligned} \lambda^\alpha &\rightarrow \Omega^{+2} \lambda^\alpha \Rightarrow \nabla \lambda^\alpha = \dot{\lambda}^\alpha + A \lambda^\alpha \Rightarrow g_h = 0 \\ \theta^\alpha &\rightarrow \Omega^{-2} \theta^\alpha \Rightarrow \nabla \theta^\alpha = \dot{\theta}^\alpha - A \theta^\alpha \Rightarrow g_h = +1 \\ \psi &\rightarrow \Omega^{-2} \psi \Rightarrow \nabla \psi = \dot{\psi} - 2A \psi \Rightarrow g_h = +2 \end{aligned}$$

BRST Trivial

* Take an operator $\mathcal{O}_{s,g}$ ghost #

Scale invariant $\sim \psi^{-s/2} \mathcal{O}_{s,g}$

$$[\psi]^{s/2} [\mathcal{O}_{s,g}] \rightarrow (\Omega^{-2} \psi)^{s/2} (\Omega^s \mathcal{O}_{s,g}) = \psi^{s/2} \mathcal{O}_{s,g} \checkmark$$

$\psi^{s/2} \mathcal{O}_{s,g} \sim T_{0,g+s} \quad g_h = g + s$

$\theta^\alpha : g_h = 1$

→ taking $\psi^{-1/2} \theta^\alpha : g_h = 1 - 1 = 0$

→ $\psi^{1/2} \lambda^\alpha : g_h = +1$

After Gauge fixing $\psi = 1$

E.O.M of A:

$$\beta = \frac{1}{2} (\omega_\alpha \lambda^\alpha + p_\alpha \theta^\alpha)$$

$$S = \int d\tau (P^m \dot{x}_m + \omega_\alpha \dot{\lambda}^\alpha - p_\alpha \dot{\theta}^\alpha) \quad (\text{Pure Spinor})$$

$$Q \theta^\alpha = \lambda^\alpha, \quad Q \omega_\alpha = -\frac{1}{2} (p_\alpha \theta^\alpha)$$

$$Q x^m = \frac{1}{2} (\lambda \gamma^m \theta), \quad Q L^\alpha = \dot{\theta}^\alpha$$